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Enhancing endurance performance with combined imagined and actual physical practice

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ABSTRACT

Endurance sports require the sustained maintenance of high effort until the point of task failure. Psychological factors, particularly the perception of effort, exert considerable influence in determining task failure. Training interventions that blend physical and cognitive tasks yielded promising results in enhancing endurance performance. Motor imagery stands out as a method capable of modulating the perception of effort. However, the precise extent to which combining motor imagery and physical training can improve endurance performance remains to be understood. In a pre-post training design, this study aimed to investigate the impact of combining motor imagery with physical training on endurance performance, compared to physical training alone. Two groups of participants were constituted (motor imagery: n = 16; control: n = 17). Both groups performed physical exercises (i.e. isometric wall squat of incremental duration, with 12 training sessions over a period of 14 days), with participants from the motor imagery group also performing motor imagery sessions. Each participant visited the laboratory for experimental procedures twice before and twice after training, during which we assessed endurance performance through a sustained submaximal isometric contraction of the right knee extensors performed until task failure (time to task failure, TTF) at either 20 or 40% of the maximal voluntary contraction. Perceptions of effort and muscle pain were measured regularly during the endurance exercise. We reported no changes in endurance performance for the control group. Endurance performance in the motor imagery group exhibited significant improvements when the intensity of the sustained isometric TTF test closely matched that used in training (i.e. 20% of MVC). However, these enhancements were less pronounced when considering higher exercise intensities (i.e. 40% of MVC). No reduction in perception of effort was observed in both groups. There was a noticeable decrease in muscle pain perception within the motor imagery group following the training. Overall, combining motor imagery and physical training may thus offer a promising avenue for enhancing endurance performance and managing pain in various contexts, from sports to clinical rehabilitation.

Keywords: motor imagery; physical training; perception of effort; perception of muscle pain; endurance performance

Impact statement

1. **Enhancing Endurance Performance Through Motor Imagery.** This study provides compelling evidence that combining motor imagery with physical training can significantly improve endurance performance, challenging conventional beliefs about the role of mental processes in endurance performance. This novel approach offers athletes and coaches a valuable tool to maximize training outcomes and enhance competitive performance.

2. Muscle Pain Reduction: A Novel Outcome. This study unveils a previously unexplored aspect of the relationship between motor imagery and physical training – a reduction in muscle pain perception. By shedding light on this phenomenon, our research offers a fresh perspective on pain management in sports and rehabilitation, potentially benefitting athletes and individuals with chronic pain or undergoing recovery.

INTRODUCTION

The ability to perform in endurance sports primarily relies on consistently maintaining the highest possible workload throughout a race. When exercising at a given workload, it is critical for athletes to postpone as much as possible the moment when they voluntarily disengage from the task, i.e. the moment when they stop exercising or decrease their level of engagement in the exercise. This upper limit of endurance is commonly referred to as "exhaustion" or "task failure".

Traditional theories in exercise sciences state that task failure is determined by physiological interactions between muscle and nervous mechanisms (Hunter et al. 2004; Allen et al. 2008; Amann et al. 2011). Recent evidences published in the last 15 years strongly challenge this assumption (Marcora and Staiano 2010; Neyroud et al. 2012; Staiano et al. 2018; Marcora et al. 2008; Souron et al. 2020; Chatain et al. 2019), and support the idea that physiological changes during endurance performance interact with psychological factors, which ultimately trigger task failure. In this context, task failure is a conscious decision to disengage from the exercise (Marcora and Staiano 2010; Marcora 2019). Inspired by the motivational intensity theory (Brehm and Self 1989; Richter et al. 2016), this perspective is developed in the psychobiological model of endurance performance (Marcora 2019; Marcora and Staiano 2010). According to this model, task failure occurs (i) when the individual perceives the effort required to perform the task as equal to or beyond her/his motivation to succeed in the task, or (ii) when the individual believes having exerted maximal effort so that exercise continuation is perceived as impossible or not worth it (Marcora 2008; Marcora et al. 2008; Marcora 2019). In this model, endurance performance is proposed to be influenced by two key factors: the motivation to succeed in the task and the perception of effort (Pageaux 2014; Marcora 2019). Specifically, any manipulation of the intensity of the perception of effort is supposed to alter endurance performance. An increased perception of effort should be associated with decreased endurance performance [e.g., effect of fatigue; Pageaux and Lepers (2016)], while a decreased perception of effort should be associated with increased endurance performance [e.g., effect of self-talk intervention; Blanchfield et al. (2014)].

Training interventions combining physical and cognitive tasks (e.g., brain endurance training) demonstrated greater effectiveness in improving endurance performance than physical training alone (Staiano et al. 2015; Marcora et al. 2013; Dallaway et al. 2021). Notably, these interventions were reported to reduce the perception of effort (Staiano et al. 2015; Marcora et al. 2013). Motor imagery is one intervention that can acutely modulate the

perception of effort (Razon et al. 2010; Razon et al. 2014), hence improving subsequent endurance performance. Motor imagery is a dynamic mental state that involves mentally representing an action without physically executing it (Di Rienzo et al. 2016). It can be performed through various sensory modes, including internal (i.e., first-person) or external (i.e., third-person) visual imagery, kinesthetic, tactile, auditory or olfactory imagery (Guillot and Collet 2008). It is widely accepted that motor imagery and actual movement execution activate overlapping neural networks across all stages of the motor pathway (Hardwick et al. 2018). This unique characteristic of motor imagery allows athletes or patients to maintain a certain level of training load or increase their training intensity by activating corticomotor pathways without producing any movement. When motor imagery is used up to ~20 minutes per training session (Jacquet et al. 2021), it has the potential to substantially enhance physical performance; e.g., muscle strength (Piveteau et al. 2023), movement accuracy (Guillot et al. 2015), movement speed (Louis et al. 2008), movement efficacy (Guillot et al. 2013a), and endurance performance (Barwood et al. 2008; Razon et al. 2010; Razon et al. 2014). Despite these promising findings, the extent to which combining motor imagery and physical training can contribute to increase endurance performance remains largely unknown.

In this context, the primary aim of this study was to investigate the effect of combining motor imagery and physical training, as opposed to physical training alone, on endurance performance. According to the proposed positive effect of motor imagery on performance, we hypothesized a greater improvement in endurance performance in participants performing the combined training. In line with the psychobiological model of endurance performance (Marcora 2019; Marcora and Staiano 2010), the secondary aim of this study was to test if the improved endurance performance would be associated with a reduction in the intensity of the perception of effort. We hypothesized that participants showing an improved endurance performance would report lower intensity of effort perception during the post-training endurance exercise compared to the pre-training endurance exercise.

METHODS

Participants. Thirty-four recreationally active participants (11 female; age: 24 ± 3 yr; height: 173 ± 8 cm; weight: 68 ± 9 kg) were recruited for this study. One participant was excluded after reporting concurrent involvement in another experimental study that required repeated sessions of intense physical training. To the best of our knowledge, no previous study investigated the effect of combining motor imagery and physical training on endurance performance. We decided to *a-priori* determine our sample size based on published studies demonstrating a positive effect of psychological interventions on endurance performance (Lakens 2022). As self-talk interventions have been shown to enhance endurance performance (our primary outcome of interest) via a reduction in perception of effort (our secondary outcome of interest), we based our sample size on these studies (Basset et al. 2022; Blanchfield et al. 2014). These studies have sample size ranging from 24 (Blanchfield et al. 2014) to 29 (Basset et al. 2022). Considering potential attrition, we recruited 34 participants, and one was excluded as previously described. Our reached sample size, according to GPower 3.1 software calculation, allows detecting a moderate effect size of f = 0.30 (i.e. η_p^2 ~0.085), with a power of 0.9, for the within-between interaction of a repeated-measures ANOVA and with correlations among measures being equal to 0.5.

All participants engaged in physical activity at least once per week. They were provided with written instructions detailing all study procedures but were unaware of its objectives and hypotheses. They were not informed that motor imagery was used to enhance endurance performance. Participants were instructed not to consume caffeine and nicotine for at least 3 h before testing and were asked to declare if they had taken any medication or had any acute illness, injury, or infection. Written informed consent was obtained from each participant. This study adhered to the standards outlined in the latest revision of the Declaration of Helsinki (with the exception of registration in a database) and was approved by the Nantes University ethics committee (n° IRB: IORG0011023).

Study design. Participants were randomly assigned to either an active control (n = 17, 12 males and 5 females) or a motor imagery (n = 16, 10 males and 6 females) group. The active control group consisted of physical training alone, and the motor imagery group consisted of combining physical training with motor imagery. The study was a controlled Pre-Post training design in which participants visited the laboratory on five separate occasions (Figure

1). Each experimental session was performed at the same time of the day for a given participant.

The first laboratory visit involved a familiarization with the procedures, completion of questionnaires, and determination of the initial performance at the isometric wall squat exercise used during the 2-week training period. After a minimum of two days, participants returned to the laboratory for two Pre-training (Pre) sessions, interspaced with at least 48 h in-between. During these sessions, endurance performance was measured with a sustained submaximal isometric contraction of the right knee extensors performed until task failure (time to task failure, TTF). We used an isometric modality of exercise for its transferability to home-based physical training sessions and its relative simplicity in assessing neuromuscular function (see below for more details). Two contraction intensities were randomly tested between Pre-sessions: 20 and 40% of the maximal voluntary contraction (MVC) peak torque. The TTF at 20% MVC peak torque ($TTF_{20\%}$) was chosen based on pilot experiments, as it had a comparable exercise duration to the wall squat posture used during training sessions (mean ± SD durations of the wall squat exercise and the $TTF_{20\%}$: 276 ± 239 and 267 ± 109 s, respectively). The TTF at 40% MVC peak torque ($TTF_{40\%}$) was chosen to investigate a potential transfer to exercise performed at a greater intensity for shorter duration. During the two TTF tests, perceptions of effort and muscle pain were measured. Knee extensors' neuromuscular function was measured before (baseline) the TTF and immediately at task failure. Two days after the second Pre-session, participants completed a 2-week training period. We opted for this specific training duration for two primary reasons: firstly, due to time constraints, as it is relatively short, and secondly, because such a brief period of motor imagery training has proven to be adequate for enhancing motor performance (Grosprêtre et al. 2018; Zhang et al. 2011). The training program involved daily wall squat exercises combined (motor imagery group) or not (control group) with motor imagery (see section *Training modalities* for more details). Two days after the completion of the training period, participants were retested in two Post-training (Post) sessions. The Pre-Post order of $TTF_{20\%}$ and $TTF_{40\%}$ was randomized between participant and kept the same for each participant.

Familiarization. After a warm-up including submaximal isometric contractions, participants performed three shorts (~3 s) isometric MVCs with the right knee extensors. Following a 3-min resting period, participants were instructed to perform an isometric wall squat exercise. They were asked to maintain the wall squat position with the knee at ~90° until task failure,

defined as the moment when (i) the participant voluntarily stopped the task or (ii) the experimenter stopped the task because the participant was unable to maintain the ~90° sitting position for more than 3 s. The experimenter noted the total exercise duration, which was subsequently used for the training program (see below). Then a 15-min resting period was given, during which participants were accustomed with all questionnaires and scale instructions. They also completed the French version of the Movement Imagery Questionnaire 3 (Robin et al. 2020). Following this resting period, participants were familiarized with the TTF tests. First, participants performed the $TTF_{40\%}$. Following a second 15-min break, participants performed the $TTF_{20\%}$. Before and immediately after the TTF tests, participants were further familiarized with neuromuscular function assessment procedures.

Experimental sessions. Each experimental session started with a neuromuscular evaluation of the right knee extensors, which included determining the baseline MVC peak torque. During and after the MVC, peripheral nerve stimulation was used to elicit superimposed and resting twitches. Participants then performed the TTF test, which involved sustaining a submaximal isometric contraction at either 20 or 40% of the MVC peak torque until reaching task failure. Task failure was defined as the point at which participants were unable to maintain the target torque for three consecutive seconds. For the Post-sessions, the MVC peak torque measured during the Pre-session was used to set the target torque for the $TTF_{20\%}$ and $TTF_{40\%}$. This ensured the investigation of endurance performance at the same absolute torque before and after the training program (control or motor imagery). Perceptions of effort and muscle pain were recorded every 20 and 30 s during the $TTF_{40\%}$ and $TTF_{20\%}$ tests, respectively. Immediately after reaching task failure, the same neuromuscular evaluation completed before the TTF was performed to assess the exercise-induced decline in torque capacities (i.e., knee extensors' fatigability) and its etiology. Standardized encouragements (e.g., "you are doing good, keep going", "do not give up", "keep going, you are doing a good job") were provided during the exercise after asking for perceptions of effort and muscle pain, in a neutral tone. If participants deviated from the target torque feedback, the experimenter calmly requested them to "return to the zone". If participants were unable to return to the target torque within 3 s, despite the experimenter's request, we considered it a task failure.

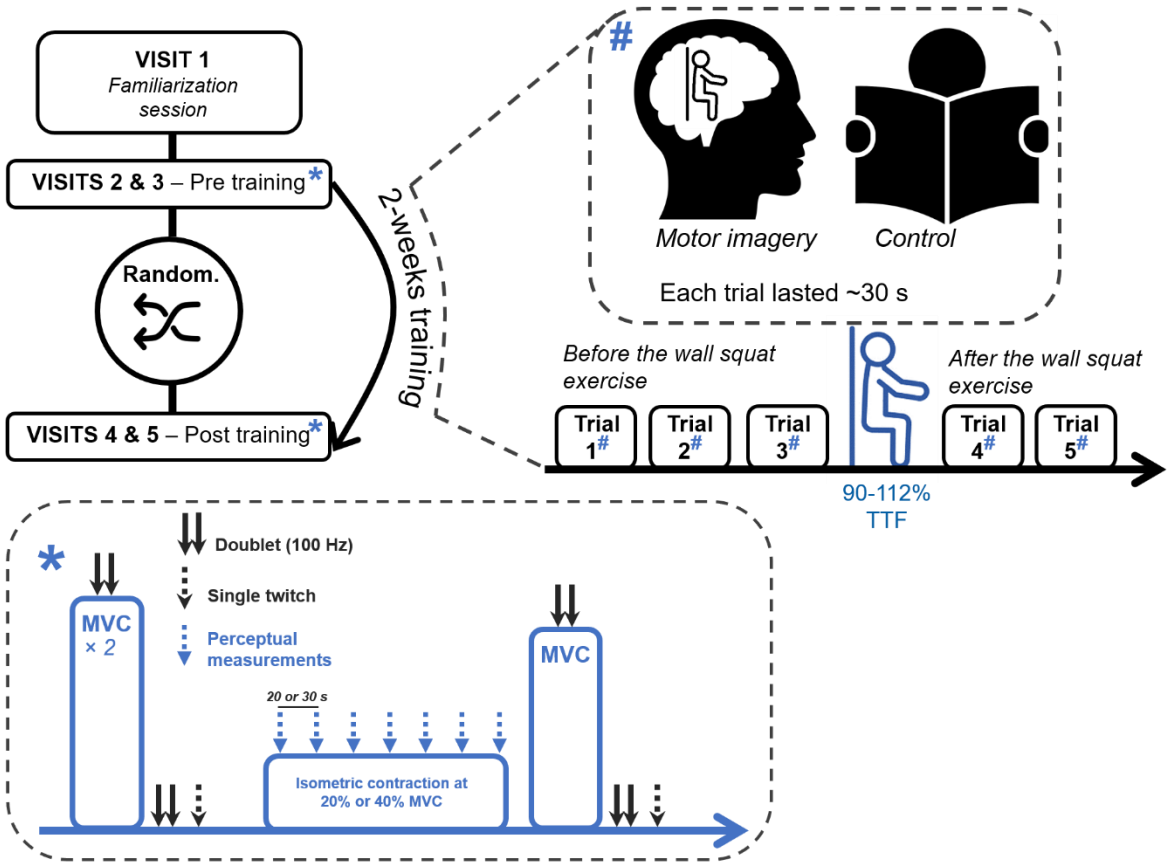


Figure 1. Overview of the experimental protocol. Each participant performed a familiarization session (visit 1) and four experimental sessions (visits 2 to 5) that consisted in performing a sustained isometric task with the knee extensors at either 20 or 40% of the maximal voluntary contraction (MVC) peak torque, until task failure. Perceptions of effort and muscle pain were recorded at regular intervals during the exercise. MVC peak torque, peripheral and central components of knee extensors fatigability were assessed before and at task failure. A two-week physical training (i.e., wall squat exercise) with (motor imagery group) or without (control group) the addition of motor imagery was performed between the third (end of Pre) and fourth (beginning of Post) visits.

Training modalities.

Both control and training groups performed physical exercises in part of their training. While the motor imagery group performed motor imagery in addition to the physical exercises, the control group performed a time-matched neutral cognitive task in addition to physical exercises (see below).

Physical exercise. Six times a week for two weeks (for a total of 12 training sessions over a period of 14 days), participants in both the control and motor imagery groups engaged in the same physical training routine at home. The training sessions consisted of performing a wall squat isometric exercise (Figure 1). During the first training session, the exercise duration was set to 90% of the total exercise duration performed during the familiarization

session. The exercise duration was gradually increased throughout the training period. Each subsequent training session incremented the duration by 2% compared to the previous session, i.e., the exercise duration for the second session was set to 92%, the third session to 94%, and so on, until reaching 112% of the initial total exercise duration. This progression scheme was chosen to strike a balance between progression within the training period and feasibility. If a participant discontinued the exercise before reaching the target duration, they were instructed not to resume the exercise, and the session was concluded. At the end of each training session, participants from both groups provided feedback on their perceived workload with the National Aeronautics and Space Administration's Task Load Index (NASA-TLX) (Hart and Staveland 1988).

Motor imagery. Participants in the motor imagery group received specific instructions from a sport psychology and motor imagery training specialist (AG). They were given standardized instructions to perform five 30-s imagined isometric contractions of the wall squat exercise during each training session. The first three imagined isometric contractions were performed just before the wall squat exercise, while the last two imagined contractions were performed immediately after the wall squat exercise, with ~30 s recovery between trials (Figure 1). Participants were prompted to combine visual and kinesthetic imagery modalities using a first-person perspective. They were asked to imagine themselves performing the final 30 s of the isometric wall squat exercise that they performed during the previous (for the three first imagined contractions) or the actual training day (for the two last imagined contractions). The focus of their imagery was on appropriately maintaining muscle contractions and body position during the wall squat exercise, disregarding any perceived fatigue. Additionally, participants were encouraged to manage and cope with discomfort and pain that might arise during the exercise. Accordingly, they were instructed to pay attention to their ability to successfully complete the wall squat exercise despite intense muscle contractions, maintaining proper body posture, and managing their effort throughout the motor sequence. Participants remained seated on a chair during the whole motor imagery procedures.

Control cognitive task. In the control group, the imagined contractions were substituted with reading from a collection of articles covering neutral topics such as sport, history, ecology and economy. If a participant did not complete reading a sentence by the end of the 30-s period, she/he was allowed to finish it to prevent any potential frustration associated with the control task.

Psychological measures

Perception of effort. Participants were instructed to report the intensity of their perceived effort using the CR100 scale (Borg and Kaijser 2006). Detailed and standardized instructions on how to use the scale and dissociate between effort and muscle pain perceptions were provided during the familiarization session (Pageaux et al. 2020). Briefly, participants were asked to rate how hard and intense it was for them to drive their leg and breathe. The CR100 scale ranges from 0 (nothing at all effort at all) to 100 (maximal effort). The maximal effort was anchored as the greatest intensity of effort ever experienced by one participant during a physical task. Modifications on the scale were made on the anchoring 0 and 100 to explicitly incorporate the term “effort” in the description (Jacquet et al. 2021). The scale included verbal anchors that correspond to different levels of intensity, such as weak/light (13), moderate (23), strong/heavy (50) or very strong (score of 70). If participants perceived an intensity of effort stronger than 100, "Maximal" (i.e., higher than the greatest effort ever experience), they were instructed that they may use numbers on the scale above 100, such as 110, 120, or even higher. Verbal examples illustrating "no effort" and "maximal effort" were provided to each participant as a reference.

Perception of muscle pain. A separated modified CR100 scale was used to measure muscle pain. The word “pain” was added in the description of the anchors 0 (no pain at all) and 100 (maximal pain). As for effort, memory anchoring was used to anchor the 100. Participants were instructed to rate “the intensity of pain they felt during the exercise, with this perception depending on localized sensations in the quadriceps (knee extensor) muscle”. These sensations were described as potentially including feelings of burning, intense or sharp pain (O'Connor and Cook 1999). It was emphasized to participants that they should not confuse or equate the rating of effort with the perception of pain (Pageaux 2016; Bergevin et al. 2023). Each participant received verbal examples illustrating "no muscle pain" and "maximal muscle pain" as reference points for using the scale.

To compare perceptions of effort and muscle pain before and after the training period, we used an “individual isotime” method (Nicolò et al. 2019). This is an efficient method to analyze the effects of an experimental manipulation and performance ranking on physiological and psychological variables measured during a TTF exercise (Nicolò et al. 2019). For each participant, we identified the shortest $TTF_{20\%}$ and $TTF_{40\%}$recorded between the Pre and Post sessions. The effort and pain perception intensities reported at the last recorded time point before task failure were considered the isotime point and were compared between the Pre- and Post-training assessments. For example, if a participant

had a TTF of 303 s during the Pre-training session and 357 s during the Post-training session, the effort and pain perception measurements at the 300-second mark were used for analysis.

Subjective workload. Participants reported their perceived workload for each training session using the NASA-TLX scale (Hart and Staveland 1988). The NASA-TLX scale consists of six items, i.e., (i) mental demand, (ii) physical demand, (iii) temporal demand, (iv) performance, (v) effort, and (vi) frustration. Participants rated each item on a scale divided into twenty equal intervals anchored by a bipolar descriptor (i.e., high/low). This score was multiplied by five, resulting in a final score between 0 and 100 for each item.

Motor imagery ability. Participants from both groups completed the French version of the Movement Imagery Questionnaire 3 [MIQ-3f; Robin et al. (2020)] during the familiarization session. The MIQ-3 is a self-report questionnaire composed of 12 items that evaluate the ability to imagine four specific movements (arm abduction and adduction, standing hip flexion, knee lift and jump) after actual physical performance, using external visual imagery, internal visual imagery, as well as kinesthetic imagery (Robin et al. 2020). Participants rated the difficulty of forming external and internal visual imagery as well as kinesthetic imagery using two 7-point scales ranging from 1 (very hard to see/feel) to 7 (very easy to see/feel). A mean score ranging from 1 to 7 was calculated for each imagery modality, with higher scores representing higher motor imagery abilities.

Motor imagery ease and vividness. At the end of each training session, participants in the motor imagery group rated the vividness of the imagined contractions and the ease of imagining these contractions using a Likert-type scale. The scale ranged from 1 (“absence of visual/kinesthetic information during imagination”) to 10 (“comparable visual/kinesthetic information as during physical practice”). Because the temporal features of the imagery task have been shown to be relevantly considered in thorough imagery quality evaluations (Guillot et al. 2012), the actual time of each motor imagery trial (that was supposed to be 30 s) was timed for the fourth, eight and twelfth training sessions during which the experimenter was connected in videoconference with the participant and timed the exact duration of each imagined contraction. The participant announced a “go” for the beginning and a “stop” for the end of each imagined contraction to help the experimenter to correctly time the duration.

Physiological measures

Torque recording. Participants sat upright on an isometric ergometer chair (S2P Ltd, Ljubljana, Slovenia) with knee and hip at 90° of flexion. The knee axis was aligned with the

axis of the dynamometer's lever arm. The minimally padded shank support was adjusted for each participant approximately 2 cm above the lateral malleolus. Hip and knee fixation was ensured with rigid straps over the pelvis and knee. Participants were instructed to push against the strain gauge (i.e., knee extension) with their right leg. The torque signal was acquired using a PowerLab data acquisition system (16/30–ML880/P, ADInstruments, Bella Vista, Australia) at a sampling rate of 2 kHz.

In each experimental session, and after a standardized isometric warm-up, participants performed two ~3 s MVCs with the right knee extensors at baseline before the sustained TTF isometric test (with the best MVC peak torque kept for statistical analysis) and one MVC immediately at task failure. The difference between Pre-Post MVC peak torques was considered as an index of knee extensors fatigability. During the MVCs and the TTF tests, the torque produced was displayed in live on a monitor position from of the participant. Strong verbal encouragements were given during MVCs.

Peripheral nerve stimulation. We used a constant-current stimulator (DS7AH, Digitimer, Hertfordshire, UK) for administering electrical stimulation to the right femoral nerve. Single rectangular electrical pulses with 0.2-ms duration and 400 V maximal output voltage were delivered via a 30-mm diameter surface cathode (Meditrace 100) taped to the skin into the femoral triangle and a 50 $\times$ 90 mm anode (Durastick Plus; DJO Global Vista, CA) in the gluteal fold. At the beginning of each experimental session, single stimuli were delivered incrementally by steps of 10 mA until resting M-wave and twitch amplitudes plateaued to determine the optimal intensity of stimulation. The optimal intensity was then increased by 20% for the electrical stimulation to be supramaximal.

Electrical stimulations were delivered in the following order: (i) a 100 Hz-doublet during the MVC, (ii) a 100 Hz-doublet on the relaxed muscle 2 s after the MVC, (iii) a single stimulation on the relaxed muscle 5 s after the MVC (Figure 1). Potentiated peak doublet and twitch amplitudes were measured via doublet and single electrical stimulations performed on the relaxed knee extensors (i.e., after the MVC), respectively, and were used as markers of muscle contractility. The ratio of the amplitude of the superimposed doublet over the amplitude of the control doublet was calculated to obtain the percentage of voluntary activation (i.e., indicative of the strength of the neural drive) as follows:

$$\text{Voluntary activation} = (1 - \text{SIT}_{\text{MVC}} / \text{peak doublet}) \times 100$$

where SIT_{MVC} is the superimposed twitch evoked during MVC.

Statistics

Statistical analyses were conducted with Jamovi 1.2.27.

Questionnaires. Independent t-tests were used to assess differences in baseline MIQ-3f global and sub scores between control and motor imagery groups. For NASA-TLX interpretation, the mean score of each item of the twelve training sessions was inputted in the statistical analysis (i.e., one general score for each item). Independent t-tests were then used to compare each item separately between groups.

Time to task failure. Independent t-tests were performed to compare TTF values of the two groups before the training intervention. Within each group, paired t-tests were performed to evaluate the effect of the two-week training period, comparing TTF values (in s) between Pre- and Post-training sessions. To answer our main research question as to whether participants from the motor imagery group improved more at Post-training (i.e., longer Post-training TTF value after adjusting for Pre-training value) than participants from the control group, we used analysis of covariance (ANCOVA) with Post-TTF as the dependent variable, Pre-TTF as the covariate and group (motor imagery vs control) as the fixed factor. ANCOVA is the method of choice for analyzing the results of trials with baseline and post treatment measurements (Vickers 2001).

Perceptual measurements. Independent t-tests were performed (i) on the first perceptual measurements that were recorded during the exercise (i.e., 3 s after the beginning of the isometric task) to confirm that both groups reported identical perceptions at the beginning of the exercise, and (ii) on the perceptual measurements recorded at task failure to confirm that both groups reported similar perceptions at exercise termination. Within each group, paired t-tests were performed to evaluate the effect of the two-week training period, comparing perceptual isotime measurement (in CR100 units) between Pre- and Post-training sessions. ANCOVA were performed with isotime perceptions of effort and muscle pain recorded Post-training as the dependent variable, isotime perceptual measurements recorded Pre-training as the covariate, and group as the fixed factor.

We used Pearson's correlation coefficient to assess the strength of the association between Pre-Post changes in both perceptual measurements at isotime and TTF performance.

Neuromuscular function and fatigability. Independent t-tests were performed to compare MVC peak torque of the two groups before the training. To assess knee extensors fatigability, we performed statistical analysis on an MVC delta, that is the MVC peak torque (in Nm) recorded at task failure minus the MVC peak torque recorded before the sustained exercise. To detect potential between-group differences in the amplitude of knee extensors

fatigability after the training period, we used ANCOVA with Post-training MVC delta as the dependent variable, Pre-training MVC delta as the covariate, and group as the fixed factor. Similar analyses were performed resting twitch and doublet amplitude, as well as voluntary activation.

All variables were normally distributed (Shapiro-Wilk test). The alpha level for statistical significance was set at $\alpha \leq 0.05$. Partial eta squared (η_p^2) are reported, and thresholds for small, moderate, and large effects were set at 0.01, 0.07, and 0.14, respectively (Cohen 1988). Cohen's d were calculated when t-tests were performed and thresholds for small, moderate and large effects were set at 0.2, 0.5, and 0.8 respectively (Cohen 1988). Greenhouse-Geisser correction to the degrees of freedom was applied when violations of sphericity were present. All data are presented as means ± standard deviation unless stated.

RESULTS

Manipulation checks

Pre-training MVC was not different between groups (motor imagery: 251 ± 75 Nm; control: 252 ± 73 Nm; P = 0.973, d = 0.011). Pre-$TTF_{20\%}$ was not different between groups (motor imagery: 250 ± 125 s; control: 285 ± 92 s; P = 0.351, d = 0.330). No baseline differences were found in $TTF_{40\%}$ between groups (motor imagery: 104 ± 36 s; control: 100 ± 29 s; P = 0.724, d = 0.123) either. Raw perceptual data are presented in Table 1.

The perception of effort recorded at the beginning of exercise (i.e., 3 s in) was similar between participants in both groups for Pre- ($TTF_{20\%}$: P = 0.621, d = 0.179; $TTF_{40\%}$: P = 0.600, d = 0.184) and Post-training ($TTF_{20\%}$: P = 0.800, d = 0.081; $TTF_{40\%}$: P = 0.456, d = 0.273) sessions. No between-group difference was observed at task failure for both Pre- ($TTF_{20\%}$: P = 0.183, d = 0.472; $TTF_{40\%}$: P = 0.975, d = 0.311) and Post-training ($TTF_{20\%}$: P = 0.161, d = 0.500; $TTF_{40\%}$: P = 0.277, d = 0.390) sessions.

The perception of muscle pain was similar between groups at the beginning of exercise for Pre- ($TTF_{20\%}$: P = 0.421, d = 0.286 and $TTF_{40\%}$: P = 0.740, d = 0.110) and Post-training ($TTF_{20\%}$: P = 0.500, d = 0.230 and $TTF_{40\%}$: P = 0.613, d = 0.184) sessions. No between-group difference was observed at task failure for both Pre- ($TTF_{20\%}$: P = 0.060, d = 0.700; $TTF_{40\%}$: P = 0.554, d = 0.214) and Post-training ($TTF_{20\%}$: P = 0.358, d = 0.328; $TTF_{40\%}$: P = 0.108, d = 0.596) sessions.

No between-groups difference was reported for each item of the MIQ-3f (P = 0.474–0,860, d = –0.240–0.060; Figure 2A). There was no difference in the score of each NASA-TLX items between groups (P = 0.301–0.831, d = –0.366–0.118; Figure 2B). Within the motor imagery group, the mean duration of the motor imagery trials of the fourth (35 ± 5 s), eighth (32 ± 5 s) and twelfth (32 ± 6 s) training sessions did not differ (P = 0.077; F = 2.840; η_p^2 = 0.179; Figure 2C). The perceived ease of imagery for these three training sessions was reported to be 6.2 ± 1.9, 5.9 ± 1.7 and 6.6 ± 2.4, while the mean global score of the twelve training sessions was 6.0 ± 1.7.

Table 1. Raw data for perceptual measurements (CR100 scale) recorded in participants from control and motor imagery groups at the beginning (i.e., 3 s) of the isometric exercise and at task failure, for Pre- and Post-training sessions.

20%MVC	Control group				Motor imagery group			
	Pre		**Post**		**Pre**		**Post**	
	Beginning	Task failure	Beginning	Task failure	Beginning	Task failure	Beginning	Task failure
Perception of effort	7 ± 5	75 ± 29	5 ± 4	80 ± 30	7 ± 4	87 ± 21	6 ± 5	92 ± 18
Perception of muscle pain	9 ± 11	88 ± 15	5 ± 4	94 ± 17	6 ± 6	97 ± 10	6 ± 6	99 ± 9
40%MVC	Control group				Motor imagery group			
	Pre		**Post**		**Pre**		**Post**	
	Beginning	Task failure	Beginning	Task failure	Beginning	Task failure	Beginning	Task failure
Perception of effort	11 ± 12	72 ± 26	11 ± 11	77 ± 24	13 ± 10	82 ± 24	8 ± 7	86 ± 21
Perception of muscle pain	10 ± 12	80 ± 22	9 ± 6	83 ± 24	11 ± 8	85 ± 23	11 ± 13	94 ± 12

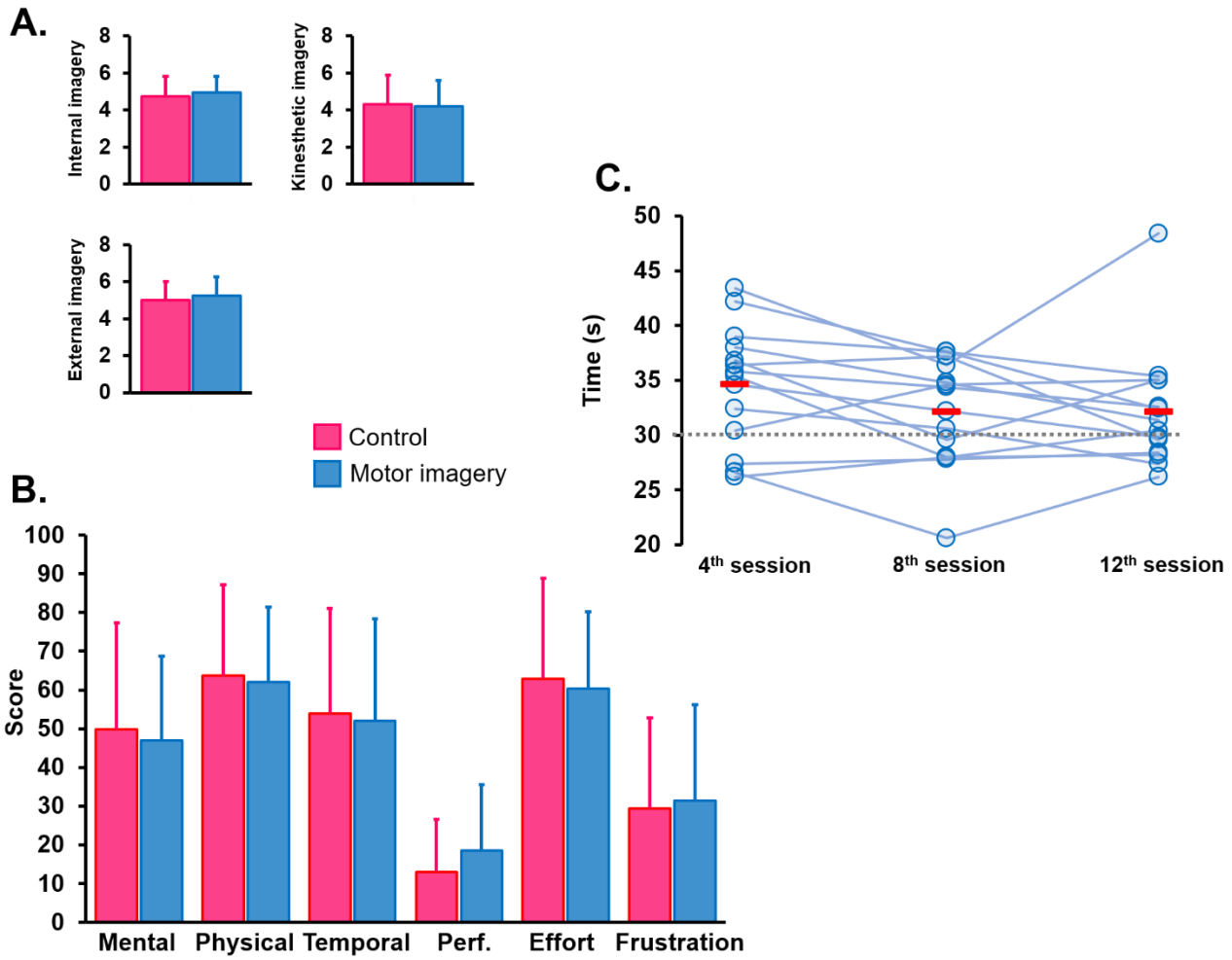


Figure 2. Panel A displays the mean (with SD) results obtained for each of the three items of the Movement Imagery Questionnaire 3f. A mean score ranging from 1 to 7 was obtained, with a higher score representing higher motor imagery abilities. Panel B displays the mean (with SD) results for the six items of the NASA-TLX questionnaire. Panel C displays the timed-duration of the imagined contractions of the fourth, eighth and twelfth training sessions for participants from the motor imagery group. Blue circles represent individual data, red line is the mean.

Effects of training on time to task failure.

$TTF_{20\%}$ increased after training for the motor imagery group (P = 0.020, d = 0.631) but not for the control group (P = 0.380, d = 0.221) (Figure 3A). The ANCOVA analysis revealed a difference among the group adjusted means for Post-$TTF_{20\%}$, with longer time to task failure in the motor imagery group (P = 0.020 F = 5.832, $\eta_p^2 = 0.160$; Figure 3B).

$TTF_{40\%}$ increased after training for the motor imagery group (P = 0.014, d = –0.683) but not for the control group (P = 0.708, d = 0.009) (Figure 3C). The ANCOVA analysis revealed no difference among the group adjusted means for Post-$TTF_{40\%}$ (P = 0.064, F = 3.753, $\eta_p^2 = 0.110$; Figure 3D).

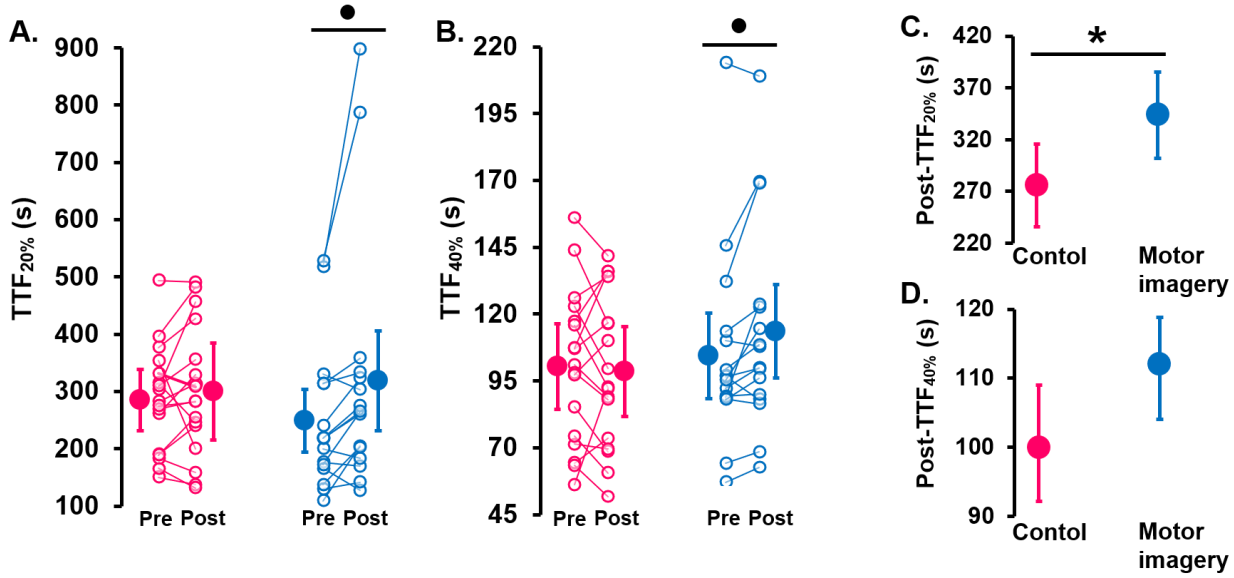


Figure 3. Performance in the isometric time to task failure (TTF) performed at 20% MVC ($TTF_{20\%}$, panels A and B) and 40% MVC ($TTF_{40\%}$, panels C and D). The control (i.e., physical training alone) and motor imagery (i.e., physical training + motor imagery) groups are presented in pink and blue, respectively. In panels A and C, data are analyzed with ANOVA, individual data are presented as empty markers and mean ± 95% confidence intervals with filled markers. In panels B and D, data are analyzed with ANCOVA, post training values are adjusted by pre-training values, and presented as ± 95% confidence intervals.• represents a significant pre-post difference within the same group, with $P < 0.05$; * represents a difference between the two groups, with $P < 0.05$.

Effects of combined motor imagery and physical training on perceptions of effort and muscle pain.

In both groups, training had no significant effect on the isotime perception of effort during the $TTF_{20\%}$ (control group: $P = 0.910$, $d = 0.355$; motor imagery group: $P = 0.104$, $d = 0.338$; Figure 4A) and the $TTF_{40\%}$ (control group: $P = 0.868$, $d = 0.280$; motor imagery group: $P = 0.573$, $d = 0.041$; Figure 4B). The ANCOVA revealed no difference among the group adjusted means for the Post-isotime perception of effort recorded during the $TTF_{20\%}$ ($P = 0.120$, $F = 2.583$, $\eta_p^2 = 0.071$; Figure 4C) and the $TTF_{40\%}$ ($P = 0.871$, $F = 0.024$, $\eta_p^2 = 0.001$; Figure 4D).

For the motor imagery group, perception of muscle pain at isotime decreased from Pre- to Post-training during the $TTF_{20\%}$ ($P = 0.040$, $d = 0.456$). There was no change in the control group ($P = 0.941$, $d = 0.404$; Figure 5A). In both groups, the training had no significant effect on the perception of muscle pain during the $TTF_{40\%}$ (control group: $P = 0.970$, $d = 0.554$; motor imagery group: $P = 0.431$, $d = 0.406$; Figure 5B). The ANCOVA revealed that adjusted Post-isotime means were different between groups during the $TTF_{20\%}$, with lower values in

the motor imagery group (P = 0.051, F = 3.900, η_p^2 = 0.110; Figure 5C). There was no difference in muscle pain during the $TTF_{40\%}$ (P = 0.260, F = 1.304, η_p^2 = 0.043; Figure 5D).

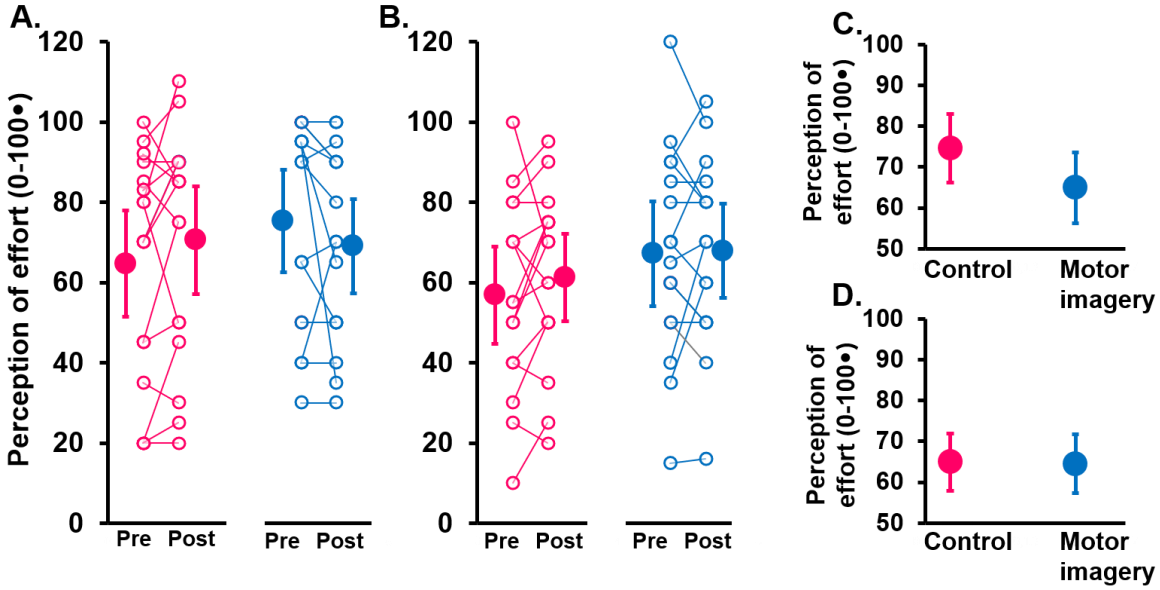


Figure 4. Pre- vs. Post-training change in the perception of effort (Borg scale, CR100) recorded at isotime during the sustained isometric contractions at 20 (panel A) and 40% (panel B) of the MVC peak torque, for control (pink color) and motor imagery (blue color) groups. Mean Pre-Post values with 95% confidence intervals are presented by the full circles. Individual Pre-Post data are represented by the empty circles. Adjusted Post-isotime means with 95% confidence intervals obtained with the ANCOVA are displayed by panels C (exercise at 20% MVC) and D (exercise at 40% MVC).

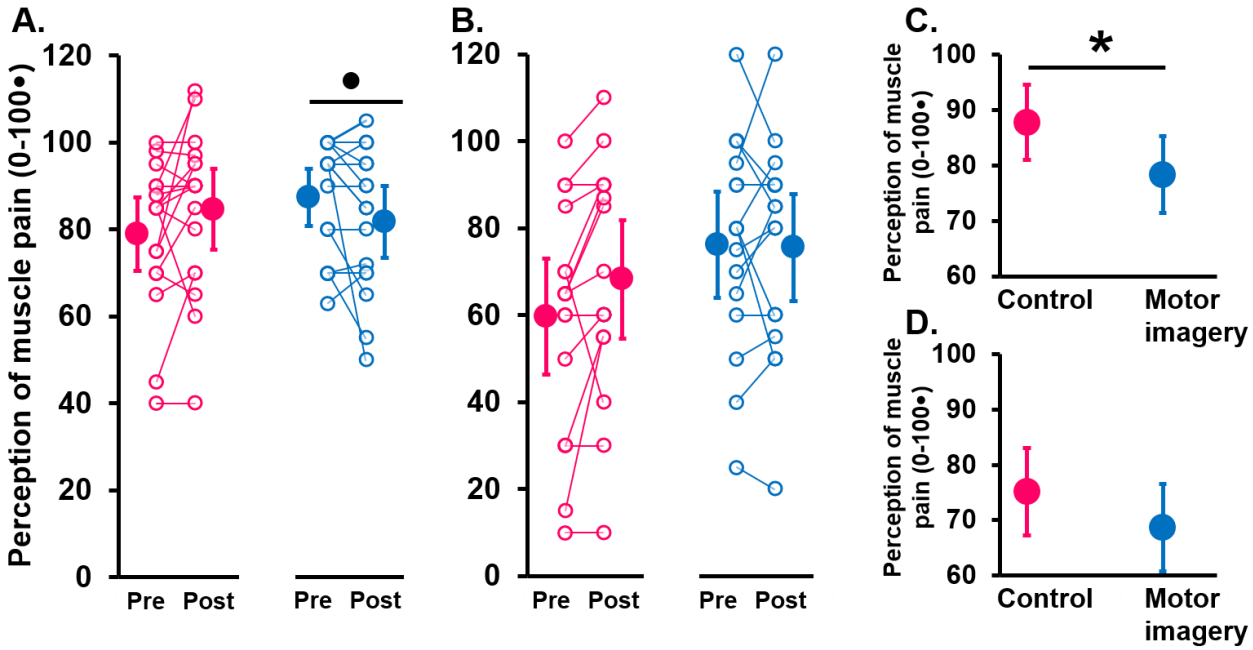


Figure 5. Pre- vs. Post-training change in perceptions of muscle pain (Borg scale, CR100) recorded at isotime during the sustained isometric contractions at 20 (panel A) and 40% (panel B) of the MVC peak torque, for control (pink color) and motor imagery (blue color) groups. Mean Pre-Post values with 95% confidence intervals are presented by the full circles. Individual Pre-Post data are represented by the empty circles. Adjusted Post-isotime means with 95% confidence intervals obtained with the ANCOVA are displayed by panels C (exercise at 20% MVC) and D (exercise at 40% MVC). .• represents a significant pre-post difference within the same group, with $P < 0.05$; * represents a difference between the two groups, with $P < 0.05$.

Relationship between changes in perceptual measurements and changes in TTF.

Pre-Post changes in both perceptions of effort (r = –0.47, P = 0.006; Figure 6A) and muscle pain (r = –0.45, P = 0.009; Figure 6B) recorded at isotime during the $TTF_{20\%}$ were correlated with Pre-Post changes in $TTF_{20\%}$ performance. The more the intensity of the perception was reduced, the greater the gain in exercise duration after the training (Figure 6).

Pre-Post changes in the perception of muscle pain were associated with Pre-Post changes in $TTF_{40\%}$ performance (r = –0.49, P = 0.004; Figure 6D), but it was not the case for the perception of effort (r = –0.18, P = 0.31; Figure 6C).

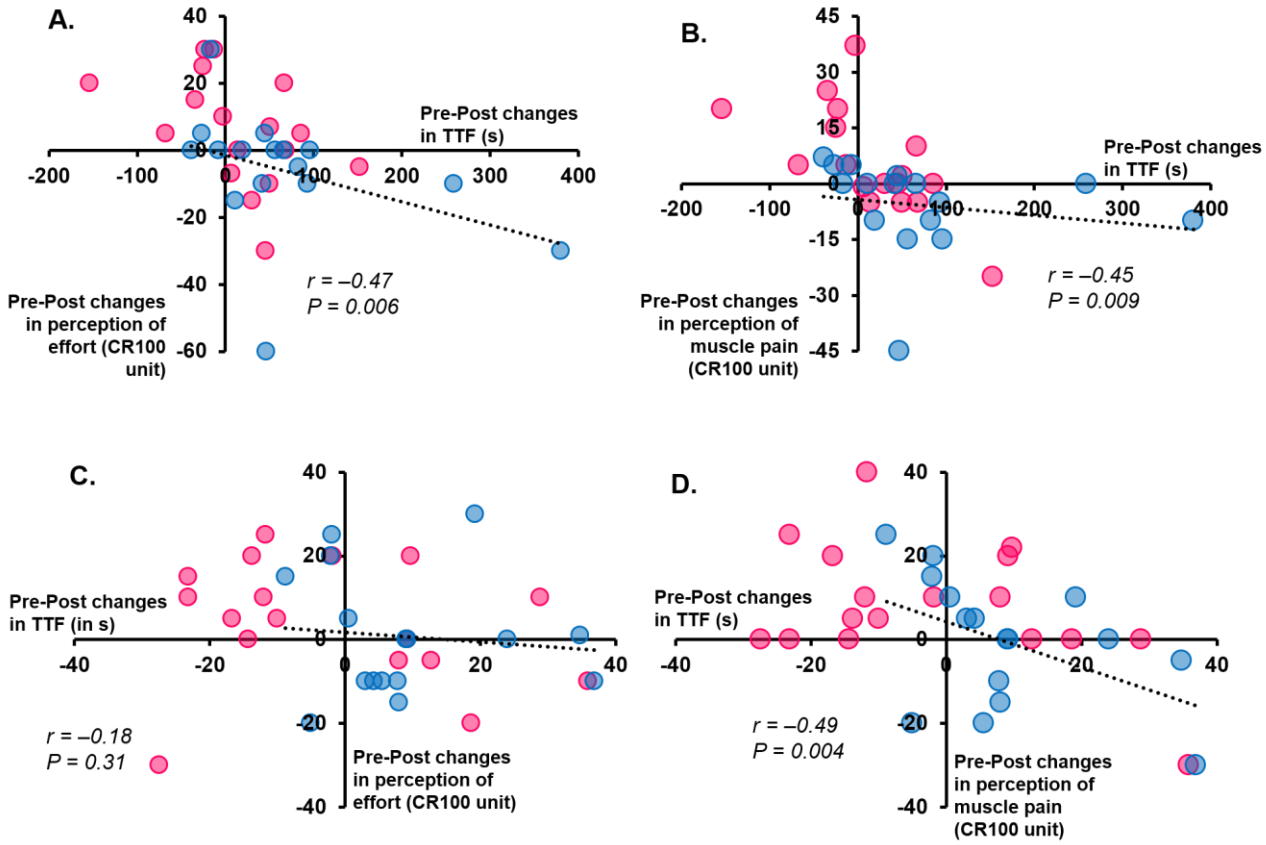


Figure 6. Panels A and B show the association between Pre-Post training changes in perceptual measurements recorded at isotime (effort and muscle pain) and Pre-Post training changes in time to task failure (TTF) for the exercise performed at 20% of the MVC peak torque. Panels C and D are for the exercise performed at 40% of the MVC peak torque. Perceptual data are Pre-Post changes and are expressed as CR100 points (negative value represents lower isotime perceptions at Post- than Pre-training). TTF data are Pre-Post changes and are expressed in seconds (positive value represents longer TTF at Post- than Pre-training). Pink and blue circles represent participants from the control and motor imagery groups, respectively.

Effects of combined motor imagery and physical training on knee extensor's neuromuscular function and fatigability

$TTF_{20\%}$. In the control group, MVC peak torque decreased by 42 ± 14% at Pre-training and 48 ± 14% at Post-training following the $TTF_{20\%}$. In the motor imagery group, MVC peak torque decreased by 39 ± 15% and 42 ± 15%. The ANCOVA analysis indicated no group effect on Post-training fatigability ($P = 0.59$, $F = 0.29$, $\eta_p^2 = 0.01$).

$TTF_{40\%}$. In the control group, MVC peak torque decreased by 41 ± 17% at Pre-training and 41 ± 12% at Post-training following the $TTF_{40\%}$. In the motor imagery group, MVC peak torque decreased by 36 ± 13% and 41 ± 12%. The ANCOVA analysis indicated no group effect on Post-training fatigability ($P = 0.34$, $F = 0.94$, $\eta_p^2 = 0.03$).

The amplitude of neuromuscular alterations was similar at Post-training between both groups for both exercise intensities. Raw neuromuscular data and associated statistics are presented in supplementary material S1.

Supplementary material S1. Raw neuromuscular data recorded in participants from control and motor imagery groups before (baseline) the isometric exercise and at task failure (TF), for Pre- and Post-training sessions. Panels A and B display the data obtained for the time to task failure tests performed at 20% and 40% of the MVC peak torque, respectively.

20% MVC	Control group				Motor imagery group				**ANCOVA analysis**		
	Pre		**Post**		**Pre**		**Post**				
	Baseline	TF	Baseline	TF	Baseline	TF	Baseline	TF	P	F	np2
MVC (Nm)	246 ± 74	144 ± 61	250 ± 66	129 ± 54	252 ± 75	153 ± 55	261 ± 74	149 ± 55	0.59	0.29	0.01
VA (%)	86 ± 8	80 ± 14	86 ± 8	77 ± 13	86 ± 8	83 ± 11	88 ± 7	82 ± 13	0.71	0.13	0.05
Peak doublet (Nm)	111 ± 31	77 ± 35	107 ± 32	73 ± 34	103 ± 31	71 ± 25	102 ± 28	67 ± 25	0.66	0.19	0.007
Peak twitch (Nm)	77 ± 24	47 ± 24	76 ± 26	44 ± 23	67 ± 23	41 ± 17	66 ± 23	38 ± 19	0.93	0.70	0.01
40% MVC	Control group				Motor imagery group						
	Pre		**Post**		**Pre**		**Post**				
	Baseline	TF	Baseline	TF	Baseline	TF	Baseline	TF			
MVC (Nm)	257 ± 73	151 ± 52	256 ± 65	152 ± 48	250 ± 76	155 ± 43	257 ± 74	150 ± 55	0.34	0.94	0.03
VA (%)	85 ± 11	87 ± 11	89 ± 6	86 ± 10	86 ± 9	86 ± 10	90 ± 5	87 ± 8	0.79	0.07	0.002
Peak doublet (Nm)	108 ± 26	66 ± 31	114 ± 30	68 ± 33	102 ± 28	63 ± 24	106 ± 29	60 ± 23	0.37	0.82	0.029
Peak twitch (Nm)	75 ± 21	38 ± 22	80 ± 23	39 ± 24	65 ± 21	32 ± 16	68 ± 24	30 ± 16	0.63	0.22	0.01

ANCOVA was performed with Post-training delta for each parameter (e.g., MVC peak torque recorded at baseline minus MVC peak torque recorded at task failure) as the dependent variable, Pre-training delta as the covariate, and group as the fixed factor.

DISCUSSION

The aim of this study was to investigate the effect of combining motor imagery with physical training, as opposed to physical training alone, on endurance performance. Contrary to the physical training alone (control) group, participants combining motor imagery with physical training improved their endurance performance. This result supports our primary hypothesis. This finding was clearly observed when the intensity of the sustained isometric time to task failure (TTF) test closely matched the one used during the 2-week training (i.e., wall squat exercise), but to a lesser extent when considering a greater intensity of exercise. Contrary to our secondary hypothesis, we did not observe a reduction in the perception of effort intensity following the combined training, despite the improved endurance performance. Interestingly, we observed a decrease of muscle pain perception.

A combined imagined and physical wall squat training, but not wall squat training alone, enhanced isometric endurance performance.

The 2-week wall squat training alone did not improve isometric endurance performance. The lack of changes in our active control group could primarily be due to the short duration of the wall squat training sessions and to the characteristics of our participants (young and recreationally active), thus preventing from the expected physiological adaptations leading to increased endurance performance. Interestingly, adding motor imagery to the wall squat training sessions led to improvements in TTF performance, specifically in the lower intensity condition (20% of MVC peak torque). Therefore, our study extends what was previously known from acute (Razon et al. 2010; Razon et al. 2014) training studies about the potential for motor imagery to improve endurance performance. Our results contribute to the growing body of evidence suggesting that a training intervention combining motor imagery and movement execution can be more beneficial than physical training alone to increase performance (Simonsmeier et al. 2021).

In the motor imagery group, the $TTF_{20\%}$ increased by 25%. This increased endurance performance was confirmed by longer TTF post training compared to control participants, with values adjusted for pre-training $TTF_{20\%}$. Results are less clear for the $TTF_{40\%}$. The 9% pre-post training increases in the motor imagery group was not different from the changes reported in the control group. By integrating these results, it seems that the greater benefits of adding motor imagery to physical training are confined to the exercise intensity closest to the one used during the combined training. As motor imagery practice induces use-

dependent neural plasticity, it is plausible that the positive effect of motor imagery on endurance performance is intensity-specific. This is supported by data showing that neural configurations at both cortical (Mizuguchi et al. 2014) and spinal (Cowley et al. 2008) levels vary based on the intensity of the imagined movement. These arguments support the notion of specific adaptations occurring in response to intensity-matched training. Therefore, it is important that motor imagery interventions include the mental rehearsal of movements closely related to the actual movements performed (Guillot 2020; Fusco et al. 2019; Guillot et al. 2013b). In our study, the intensity of the isometric wall squat exercise was similar to the one of the $TTF_{20\%}$. It is likely the reason why improvement in endurance performance in the motor imagery group was clearly observed only at 20%, and not at 40% of the MVC peak torque. Importantly, no changes in neuromuscular function were observed from pre to post training. Therefore, improvement in force production capacity of the working muscles cannot be considered as the underlying cause of the enhanced endurance performance.

No clear effect of the combined training on the perception of effort.

To understand how the combination of physical training and motor imagery could improve endurance performance, we decided to use the framework of the psychobiological model of endurance performance (Marcora et al. 2008; Pageaux 2014). This model predicts that any intervention improving endurance performance should be associated with a reduction in the perception of effort reported by the participants. Contrary to our secondary hypothesis, while we observed an improved endurance performance in the motor imagery group, we did not observe a significant decrease in the intensity of the perception of effort after the 2-week training. This observation contrasts with previous studies demonstrating that psychological (Blanchfield et al. 2014) and combined (Staiano et al. 2015) training interventions improve endurance performance though a reduction in the perception of effort. Interestingly, we still observed a medium to large correlation between increased TTF in the 20% MVC condition and reduction in the perception of effort. This correlation was not present in the 40% MVC condition where the increased performance was unclear. The discrepancy in our results, as well as the difference with the existing literature, suggest that future studies should replicate our design and increase the sample size to provide clearer conclusions on the absence or presence of reduced perception of effort induced by combining physical training and motor imagery intervention. Such studies will contribute to challenge existing models placing perception of effort as the key regulator of endurance performance, e.g. the psychobiological model of endurance performance (Marcora and Staiano 2010), the flush model (Millet 2011), and contribute to better understand the regulation of endurance performance.

673 Another possibility to explain the lack of significant reduction in the perception of effort is the
674 variability in the rating of perception of effort provided by our participants. Seven out of 33
675 participants reported having exerted a strong effort or below (values ≤ 55) at task failure,
676 which is far below the maximal or near maximal values reported by the other participants.
677 These values are also much lower than values reported in the literature at task failure in
678 isometric endurance exercises (Souron et al. 2020; Pageaux et al. 2013; Froyd et al. 2016).
679 These submaximal values are likely a result of the anchoring method we used for maximal
680 effort; i.e., the greatest intensity of effort ever experienced by one participant during a
681 physical task. It is therefore likely that these seven participants may have experienced much
682 more intense engagement in physical tasks in their past, such as performing a marathon or
683 an incremental running/cycling test until exhaustion. Such past experience would be
684 associated not only with the strong muscular solicitation that our participants experienced
685 during the knee extensor TTF, but also with strong respiratory solicitations as it is classically
686 the case during whole-body exercise. This complements the recent study of Malleron et al.
687 (2023) reporting a large difference in perceived effort ratings when participants are free to
688 select the reference for their maximal anchor compared to an imposed maximal anchor
689 related to the performance task of interest. As effort is related to a specific goal, effort is
690 therefore task specific (Bergevin et al. 2023; Pageaux 2016; Halperin and Vigotsky 2023).
691 We suggest that future studies monitoring perception of effort should impose the maximal
692 anchor in relation to the primary task of interest. For studies interested in using self-selected
693 maximal anchors, we propose that researchers carefully monitor for each participant (i) the
694 self-selected reference of the maximal anchor as well as (ii) the participant's physical activity
695 and sport background to contrast with the self-selected maximal anchor. Such approach
696 could help to better understand inter-individual differences in the perception of effort.

697

698 The *combined training decreases the perception of muscle pain.*
699 In line with the proposed role of muscle pain in the regulation of endurance performance
700 (O'Connor 2021; Zhang et al. 2023; Stevens et al. 2018), we also monitored the perception
701 of muscle pain for exploratory analysis. The exploratory monitoring of muscle pain is also
702 justified by previous observation suggesting a positive effect of motor imagery on pain
703 (Javdaneh et al. 2021; Polli et al. 2016; Volz et al. 2015). Interestingly, the addition of motor
704 imagery to physical training reduced the perception of muscle pain. The instructions
705 provided to the participants for performing motor imagery as well as the nature of the
706 endurance exercise are likely to explain this result.

Regarding the instructions provided to the participants, the motor imagery sessions were supposed to reproduce the last 30 s of the actual exercise, when muscle pain is at its highest (e.g., "*you will only focus on the last 30 s of the wall squat exercise, when the sensations of muscle pain, discomfort, and difficulty are high*"). Therefore, participants were frequently exposed to pain-related sensory experiences during the wall squat training. It is likely that this frequent experience or rehearsal of muscle pain sensation could yield to habituation effects, leading to increased tolerance to muscle pain sensations and resulting in improved isometric endurance performance. Future neurophysiological studies are needed to unravel the potential specific neuroplastic changes induced by the combination of physical training and motor imagery (Lotze and Moseley 2022; Ruscheweyh et al. 2011; Volz et al. 2015).

Regarding the endurance exercise that we selected, its isometric nature is likely to explain the key role of muscle pain in our results. The mechanical compression during a sustained isometric contraction is unremitted, which inevitably limits the local blood flow (Lind 2011), even for contractions as low as 20% of MVC peak torque (Barcroft and Millen 1939). Furthermore, contrary to dynamic exercise, isometric exercise does not activate the muscle pump, thus further contributing to the fast building of exercise metabolites (Ahlborg et al. 1972). This accumulation of metabolites within the muscle milieu, combined with the mechanical stress caused by the sustained isometric contraction, do not only contribute to the reduction in muscle force production capacity, but also to the stimulation of muscle nociceptors [afferent type III and IV fibers or Aδ and C, respectively; O'Connor and Cook (1999)]. For the reasons previously mentioned, it is likely that the pain experienced during an isometric exercise is greater than during a dynamic exercise. Future studies should attempt to replicate our results with dynamic exercises, both in the context of isolated or whole-body exercise where the cardiorespiratory solicitation differs (Sidhu et al. 2013). We also would like to emphasize that given the exploratory nature of our analysis on muscle pain, the reproduction of our results is of particular importance, with study focusing especially on this variable with specific *a-priori* hypothesis.

CONCLUSIONS

Our study offers preliminary evidence that combining motor imagery with physical training is efficient to improve isometric endurance performance. While the effect of the combined training on the perception of effort is unclear, our results highlight a reduction in muscle pain. The combined physical plus motor imagery training appears to be equally effective as other psychological interventions [e.g., Blanchfield et al. (2014); Barwood et al. (2015)] in

improving endurance performance. Of particular interest is that 12 training sessions over a span of two weeks was efficient in improving endurance performance in our population. Our findings also revealed a specific effect of imagery training, confined within the same exercise nature and intensity as the actual training performed. The positive effect of combining motor imagery with physical training beyond the intensity used in the actual training remains to be determined. The lack of transfer towards the specific exercise intensity used in the training suggests the involvement of specific neuroplastic modulations. We encourage researchers to explore this possibility as our study was not designed for this purpose. Even though the present observations need to be reproduced with athletes or individuals living with medical conditions, our results have potential practical implications in various domains, including sports and clinical settings. The integration of motor imagery training alongside physical training could enhance endurance performance in athletes, while also providing a means to manage and alleviate the perceived sensation of muscle pain. The combined training approach may also be particularly beneficial for individuals with pain conditions (e.g., chronic pain, critically ill patients) or those undergoing rehabilitation, as it offers a non-pharmacological and potentially effective method for pain management and motor performance improvement.

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